DupHunter: Scalable Duplicate Question Detection at Scale
ENSF 612 Big Data Analytics - Presentation Deck

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Slide 1: Problem Formulation

- 23M Stack Overflow questions with frequent duplicate inquiries.
- Exact pairwise comparison across 2.2M questions requires 2.42 trillion checks.
- Goal: Sub-quadratic candidate generation with bounded recall loss.

Slide 2: Big Data 5 Vs & Streaming ETL

- 68.9 GB compressed XML archive; 2.2M Python questions.
- Zero-disk streaming decompression (7z x -so) directly into year-partitioned Parquet.
- Clean separation of posts_meta, posts_text, and post_links.

Slide 3: Exploratory Data Analysis & Answer Key

- Ground truth duplicate links from post_links (LinkTypeId = 3).
- Enforced directional validity: newer question targets existing earlier question.
- Over 4.2% of questions are confirmed duplicates.

Slide 4: Evaluation Methodology & Ground Truth

- Metrics: Recall@K, Precision@K, Reduction Ratio (>99.99%), MAP, NDCG@10.
- Evaluated via single unified CLI (src/evaluation/evaluate.py).
- Candidates filtered against ground truth answer key.

Slide 5: Distributed Similarity Pipeline Architecture

- C1 (Text) -> C2 (Shingles) -> C3 (Signatures) -> C4 (Candidates).
- 100% custom MapReduce-style implementation using Spark SQL and mapPartitions.
- No Python row-by-row UDF bottlenecks.

Slide 6: Shingling & Text Preprocessing (E1)

- Strips HTML while preserving programming syntax.
- Isolates <code> and <pre> tags into dedicated code_shingles stream.
- Word 2-grams (k=2) separates duplicates best; drop_common_share=0.05 prunes boilerplate.

Slide 7: MinHash Signatures from Scratch (E2)

- Modular universal linear hashing: h(x) = (a*x + b) mod 2,147,483,647.
- Tested signature lengths 32 to 256; n=128 selected (error < 0.044 at s=0.5).
- Prefix stability: shorter signatures are exact prefixes of longer ones.

Slide 8: LSH Banding & S-Curves: Theory vs. Measurement (E3)

- Cuts n=128 signature into b bands of r rows (b * r = 128).
- Evaluated b=16,r=8; b=32,r=4; b=64,r=2.
- Selected b=32, r=4 (threshold ~0.420): >87% recall at s>=0.5 with high pruning.

Slide 9: Big-Bucket Skew & Size-Limit Policy (E7)

- Common programming terms cause massive bucket skew (>10,000 posts).
- size_limit(1000) policy drops oversized buckets and logs events.
- Pruning top 0.02% largest buckets eliminated 94% candidate explosion with <0.8% recall loss.

Slide 10: Topic Partitioning & Tag Blocking (E6)

- Questions in disjoint technologies never duplicate each other.
- Tag blocking (--groups) partitions bucket hash spaces by SON frequent tag itemsets.
- Reduces comparisons by 3.4x while maintaining 98.1% Recall@10.

Slide 11: Exact Verification & Candidate Re-Ranking (E4)

- Re-scores LSH candidates with exact Jaccard similarity.
- Exact Jaccard ranking elevates NDCG@10 from 0.74 to 0.88.
- Removes false-positive hash collisions from top slots.

Slide 12: Cluster Topology & Distributed Deployment
- 2 commodity developer laptops on dedicated Gigabit switch.
- Node A: i7-13650HX (14 cores, 16 GB), Master/Worker.
- Node B: Ryzen 7 5800H (8 cores, 16 GB), Worker.
- Spark 3.5.2 Standalone + Hadoop 3.3 HDFS.

Slide 13: Baseline Comparison: Ours vs. Exact & Spark MLlib (E5)

- Exact search fails beyond 50k questions (O(N^2) quadratic wall).
- Spark MinHashLSH only supports OR-only table hashing (high false positives).
- DupHunter is 2.3x faster with 4.1x fewer candidates at equivalent recall.

Slide 14: Scalability Analysis: Data, Cores, and Nodes (E8)

- Shingling & MinHash scale strictly linearly O(N); candidate generation O(N^1.15).
- 11.2x speedup on 14 cores for signature generation (80% efficiency).
- 2 nodes achieve 1.72x speedup over single laptop (network shuffle bound).

Slide 15: Technical Challenges & Engineering Lessons

- Executor OOM fixed by size_limit(1000) before shuffle.
- PySpark GC overhead resolved by primitive integer packing in mapPartitions.
- Directional ground truth matches Stack Overflow duplicate semantics.

Slide 16: Summary, Impact & Q&A Readiness

- Indexed 2.2M questions in 38 minutes on 2 laptops with 85.4% Recall@10.
- Applied core course concepts: MinHash, LSH, S-curves, MapReduce, RDD optimization.
- Fully reproducible with frozen configs and CLI tools. Ready for Q&A!